Warning: This task has a 10-minute time limit for submitting your solution. Once you request an input, the timer will start automatically, and after the deadline it will no longer be possible to submit a solution for that input. *You can always request a new input*, so don't worry if the timer expires: you will simply need to request and download a new input.

To help you with this task, we have prepared some ***solution templates****, which include only the input reading and output writing parts (from keyboard and to screen). You can choose to read/write from files by uncommenting the appropriate lines of code.*

- *Download the C template: taxi.c*
- *Download the C++ template: taxi.cpp*
- *Download the C# template: taxi.cs*
- *Download the Go template: taxi.go*
- *Download the JavaScript template: taxi.html*
- *Download the Java template: taxi.java*
- *Download the Pascal template: taxi.pas*
- *Download the Python template: taxi.py*
- *Download the VisualBasic template: taxi.vb*

Problem Description

The museum on the history of the *Italian Olympiad in Informatics* has just completed its inauguration, achieving a fair amount of success. Now the N visitors are waiting to take a taxi to go back home, and a long queue has formed outside the museum.



In the taxi queue, the N visitors are divided into groups. Each group is identified by its own number; the group identifiers can be any numbers, not necessarily in

any particular order.

In particular, person i belongs to group G_i, and the groups have lined up so that all people from the same group are consecutive.

Since each group wants to travel in a single taxi so as not to split up, and does not want to share the taxi with other groups, how many taxis are needed to take all visitors home?

Input Format

The first line of the input file contains an integer T, the number of test cases. Then follow T test cases, numbered from 1 to T. Each test case is preceded by a blank line.

Each test case is structured as follows:

- one line containing the integer N.
- one line containing the N integers $G_0, \ldots, G_{N-1}$.

Output Format

The output file must contain the answers for the test cases you managed to solve. For each test case you solved, the output file must contain a line with the text "`Case #test:`", where `test` is the test case number (starting from 1), followed by the integer `ans`.

Constraints

- $T = 20$, and the input files you will download will contain exactly 20 test cases.
- $1 \leq N \leq 10^4$.
- $0 \leq G_i \leq 10^4$.

Test Case Groups

The test cases are divided into groups.
Each group is worth 5 points, and to obtain the score for a group it is necessary to correctly solve all the test cases it contains.

Some groups have additional assumptions:

`Assumptions`	`Groups`
All G_i have the same value	1
G_i takes at most 2 distinct values	2
$G_i = G_{i-1}$ or $G_i = G_{i-1} + 1$ for every $i = 1 \ldots N-1$	3
No additional assumptions	$4, 5, 6, 7, 8, 9, 10$

Input/Output Examples

Input:

```
2

3
2 2 1

7
2 2 3 3 3 4 4
```

Output:

```
Case #1: 2
Case #2: 3
```

Explanation

In the **first sample case**, there are two people in group 2 and one person in group 1. Therefore, we need two taxis.

In the **second sample case**, there are two people in group 2, three people in group 3, and two people in group 4. Therefore, we need three taxis.